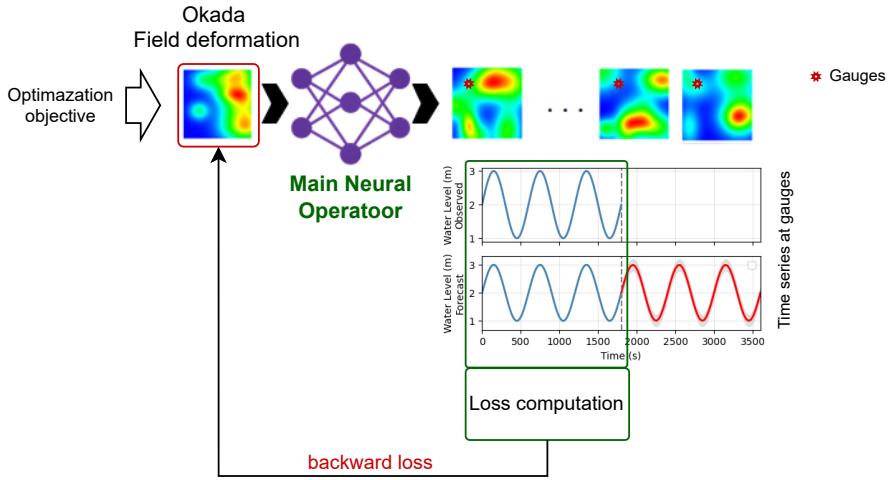
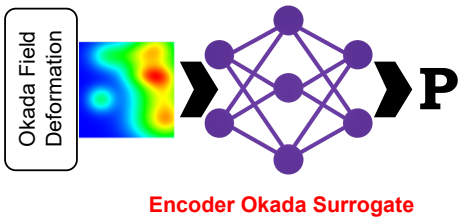
Stage 1 – Field-Space Inversion of Okada Deformation

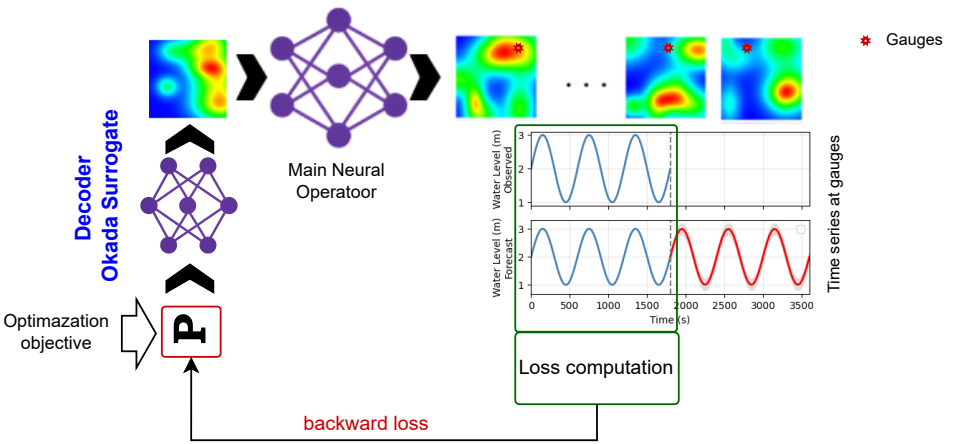
(Direct optimization of the seafloor deformation field using early gauge observations.)



Stage 2 – Parameter Extraction via Okada Surrogate Encoder

(Encoding the inferred deformation field into physical fault parameters using the learned inverse model.)



Stage 3 – Parameter-Space Inversion and Refinement

(Starting from Stage-1 initialization, optimizing Okada parameters through differentiable decoding and gauge-driven loss feedback.)